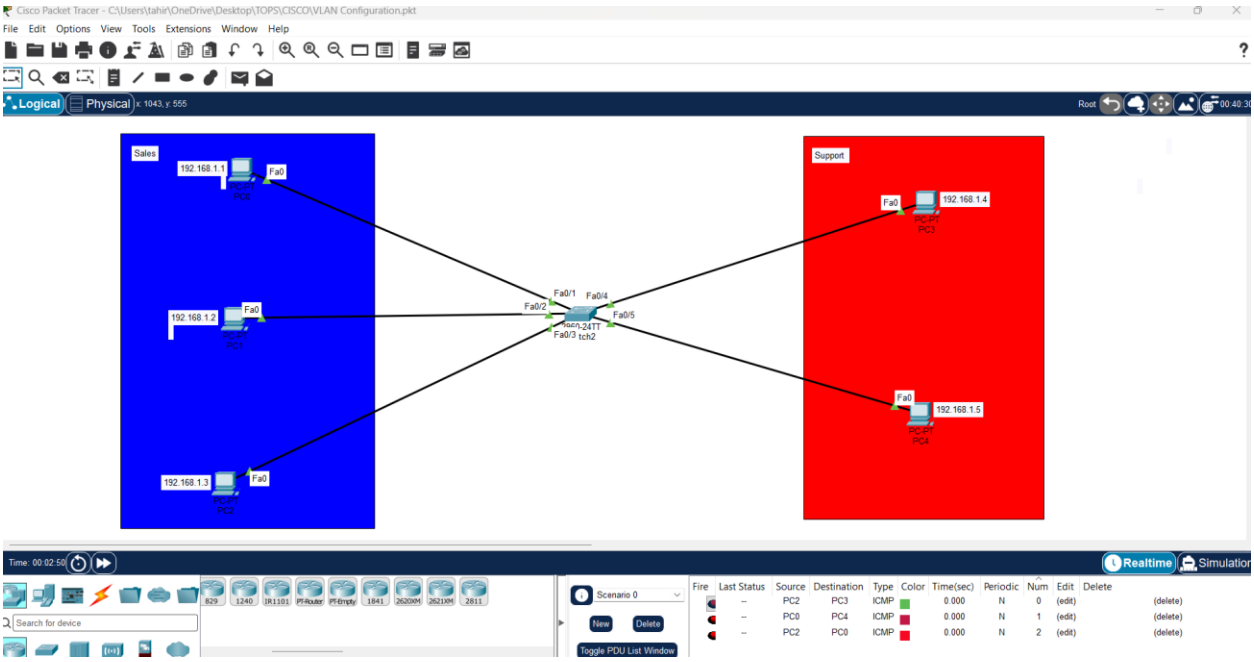
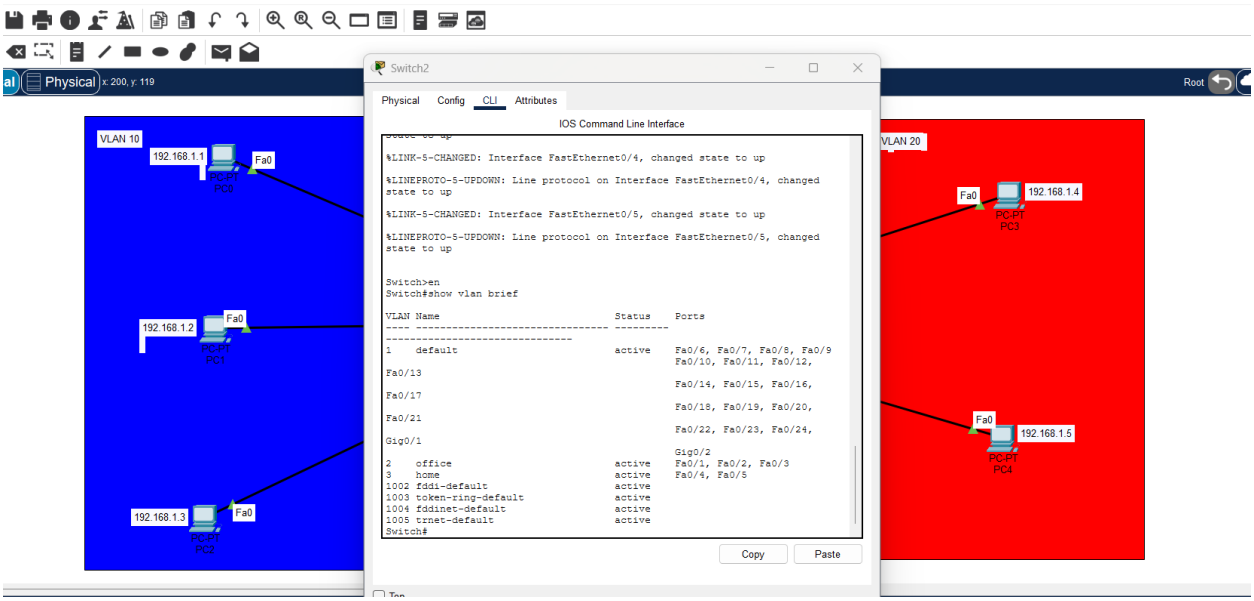
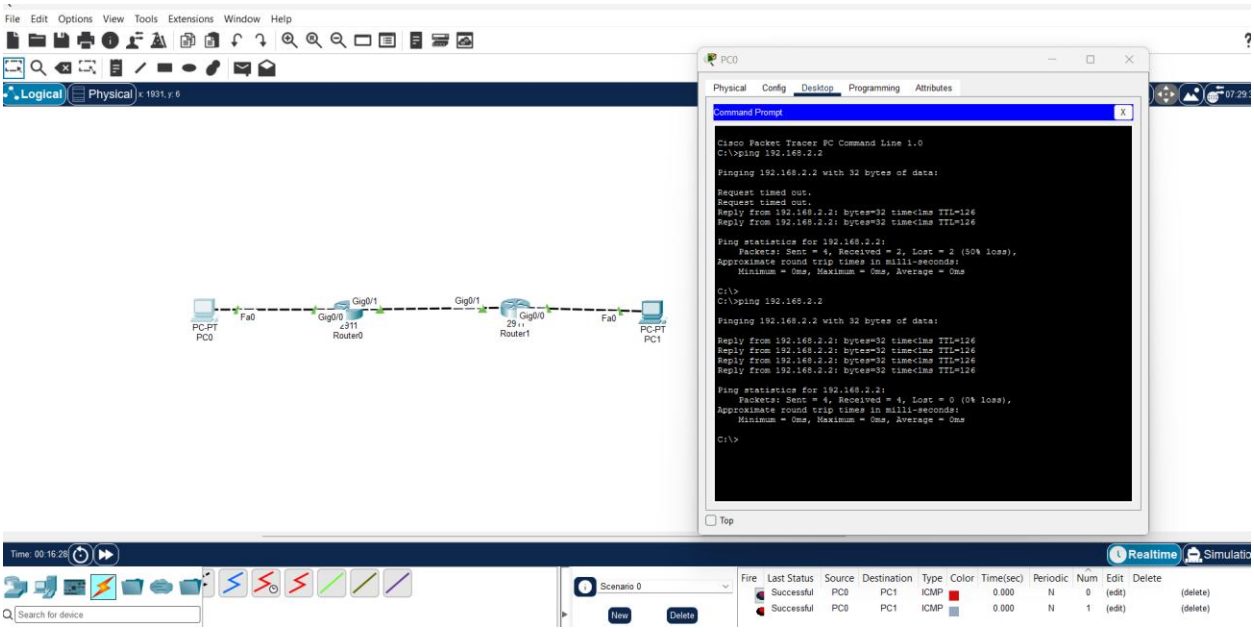
1.Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.



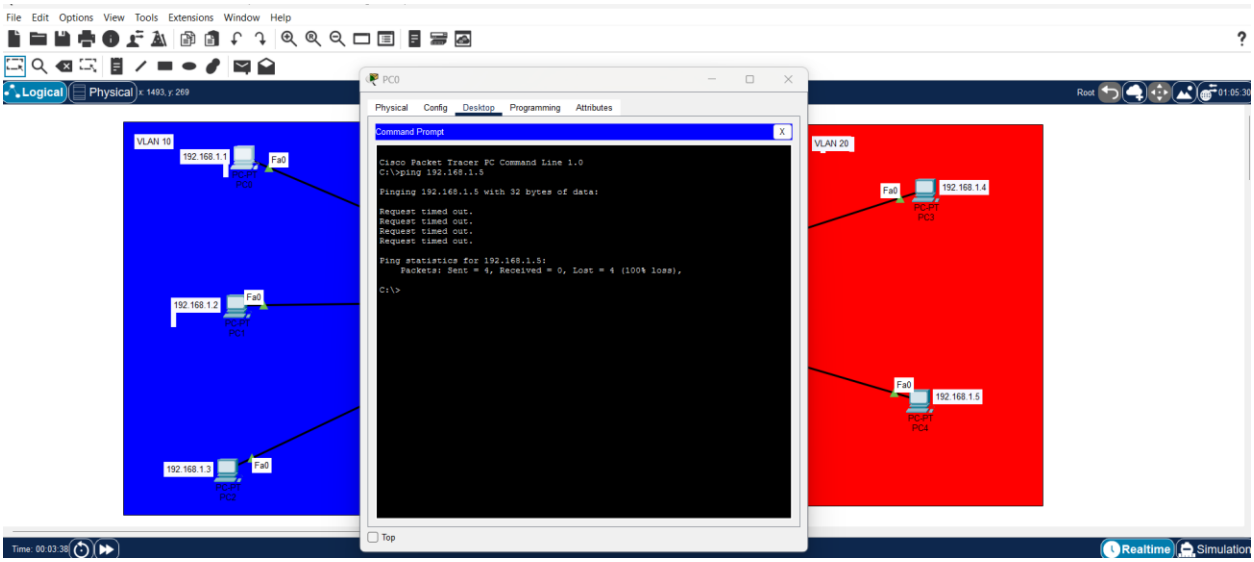
2.Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command. Hint: Use the interface range command to select multiple ports efficiently.



3.Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.



4.Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication. Hint: Think about router-on-a-stick or Layer 3 switch concepts.



Why does the ping fail?

VLAN 10 (Sales) and VLAN 20 (Support) are separate networks. A Layer 2 switch cannot send data directly between them.

How can we fix it?

We can use Router-on-a-Stick by connecting a router to the switch and configuring sub-interfaces for both VLANs. Alternatively, we can use a Layer 3 switch with routing enabled.

In simple words: Devices in different VLANs cannot communicate directly. We need a router or Layer 3 switch to allow communication between them.